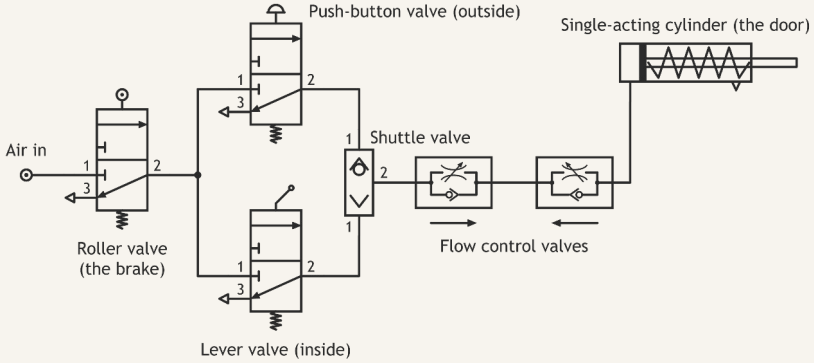
Daily Review

1	Fault-finding: check 1, every time?	Regulator at 0
2	Write up a fault: the three things?	Fault, cause, fix
3	Single-acting: what brings the rod back?	A spring
4	Which valve slows the rod?	Flow control valve
5	A sensor on a system diagram is the…	Input

The Train Door

Two ways in, one brake: can air decide?



S3 Engineering Science

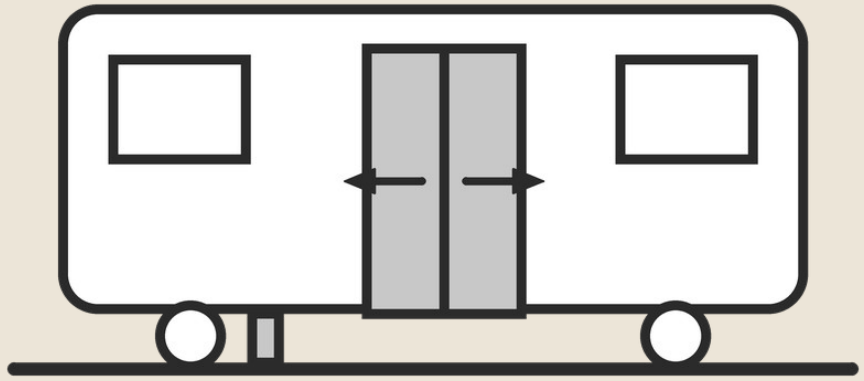
The Job: The Train Door

i When the brake is on, the button opens the door slowly.

ii When the brake is on, the lever opens the door slowly.

iii When the brake is off, the door stays shut.

Predict: brake off, lever pushed. Does the door open?



By the end of this lesson…



1. I can build a 7-part circuit from a bench card
2. I can test every way in with a table
3. I can set two flow controls to one slow speed

SAFETY: The Roller Rule

Never

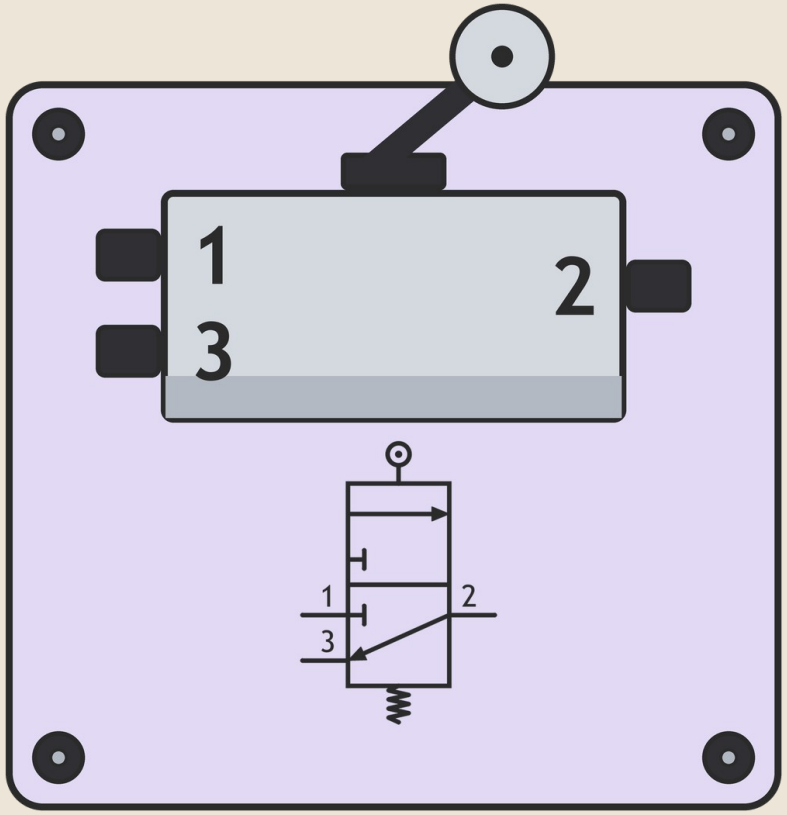
move or bolt a roller with air on

You may

press it to test

Why

it can move the door



SAFETY Air off to move a roller. Test it with one finger.

THE BUILD

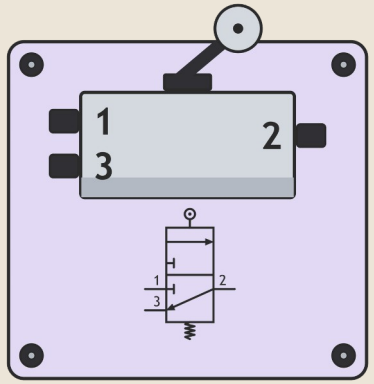
Build the Train Door

Simulate it. Air Permit. Then the kit.

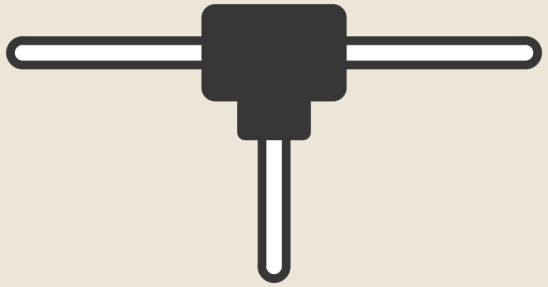
What to Collect

Builder and Safety: one set per team.

1 Bench card 5, tubing
2 Manifold (air supply)
3 Roller valve
4 Button, lever valves
5 T-connector, shuttle
6 2 flow control valves
7 Single-acting cylinder
8 Goggles, one each



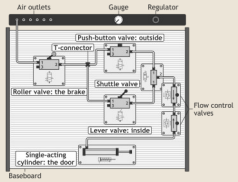
Roller valve: today, the brake.



T-connector: one in, two out.

Simulate and Build

Tasks 1-2 (B6A-L5-p15-16)

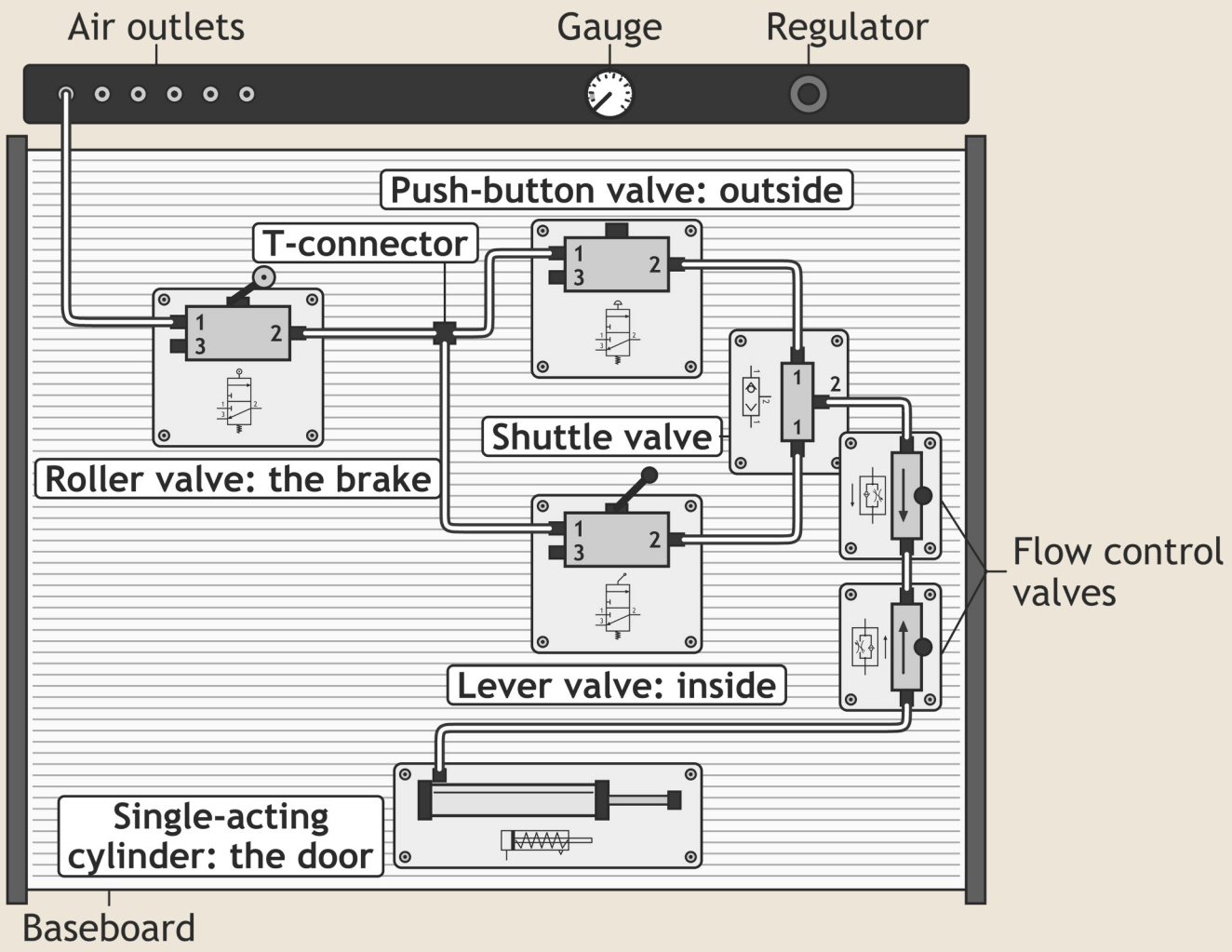


Tester and Recorder: start on the computer.
Builder and Safety: parts and bench card 5.
Safety, then teacher: check the Air Permit.

SAFETY Bolt the roller down first, with the air off.

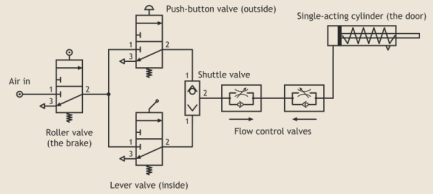
19 min

Build It: Bench Card 5



Test Every Way In

Tasks 3-4 (B6A-L5-p16)



Task 3: compare the sim and the kit.

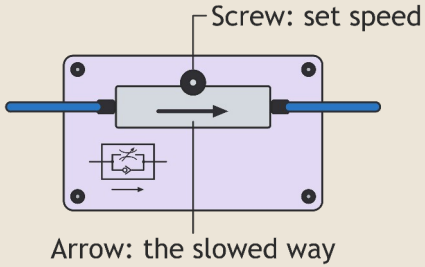
Task 4: test each row. Write open or shut.

SAFETY Press the roller with one finger. Never move it.

6 min

Beat the Clock

Task 5 (B6A-L5-p17)



Set both flow controls: open and close at the same slow speed.
Circle your target: Bronze, Silver or Gold.

SAFETY Turn a flow control only with the rod still.

4 min

Test It: What Did You Find? (B6A-L5-p16)

Job	Brake (roller)	Button (outside)	Lever (inside)	Door
iii	Off	Pressed	-	Shut
i	On	Pressed	-	Open
ii	On	-	Pushed	Open
-	On	-	-	Shut
iii	Off	-	Pushed	Shut

Plenary: How Did It Decide?

1	Brake off: where did the air stop?	At the roller valve
2	Which part did the air pass first?	The roller (the brake)
3	Button or lever: which opened it?	Either one
4	What made the door move slowly?	The flow controls
5	Move a roller only with…	The air off

Look back — by the end of this lesson…



1. I can build a 7-part circuit from a bench card
2. I can test every way in with a table
3. I can set two flow controls to one slow speed

LOOKING AHEAD

Lesson 8: How Does the Door Decide?

Inside the shuttle valve: OR, AND and the brake.

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Real part drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Train door circuit: The train door circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 3 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

R Stewart - October 2026